PRACTICAL-02

TASK-01

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

a) **Write a C code to create a process using fork() system call.**

b) **Read the user input as shell command.**

c) **Use a system call exec() to execute the shell command.**

d) **Use a wait() system call for waiting a parent process.**

e) **Display the PIDs of parent and child processes.**

```
root@SRIVALLI:~# nano command_executor.c
root@SRIVALLI:~# gcc command_executor.c -o command_executor
root@SRIVALLI:~# ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID: 557
Child PID : 558

Child Process
Child PID : 558
Parent PID: 557

Executing command: ls
OSSP                command_executor.c.save  nano.443.save
command_executor    fork_process             process_state
command_executor.c  fork_process.c           process_state.c

Child process completed.
root@SRIVALLI:~#
```

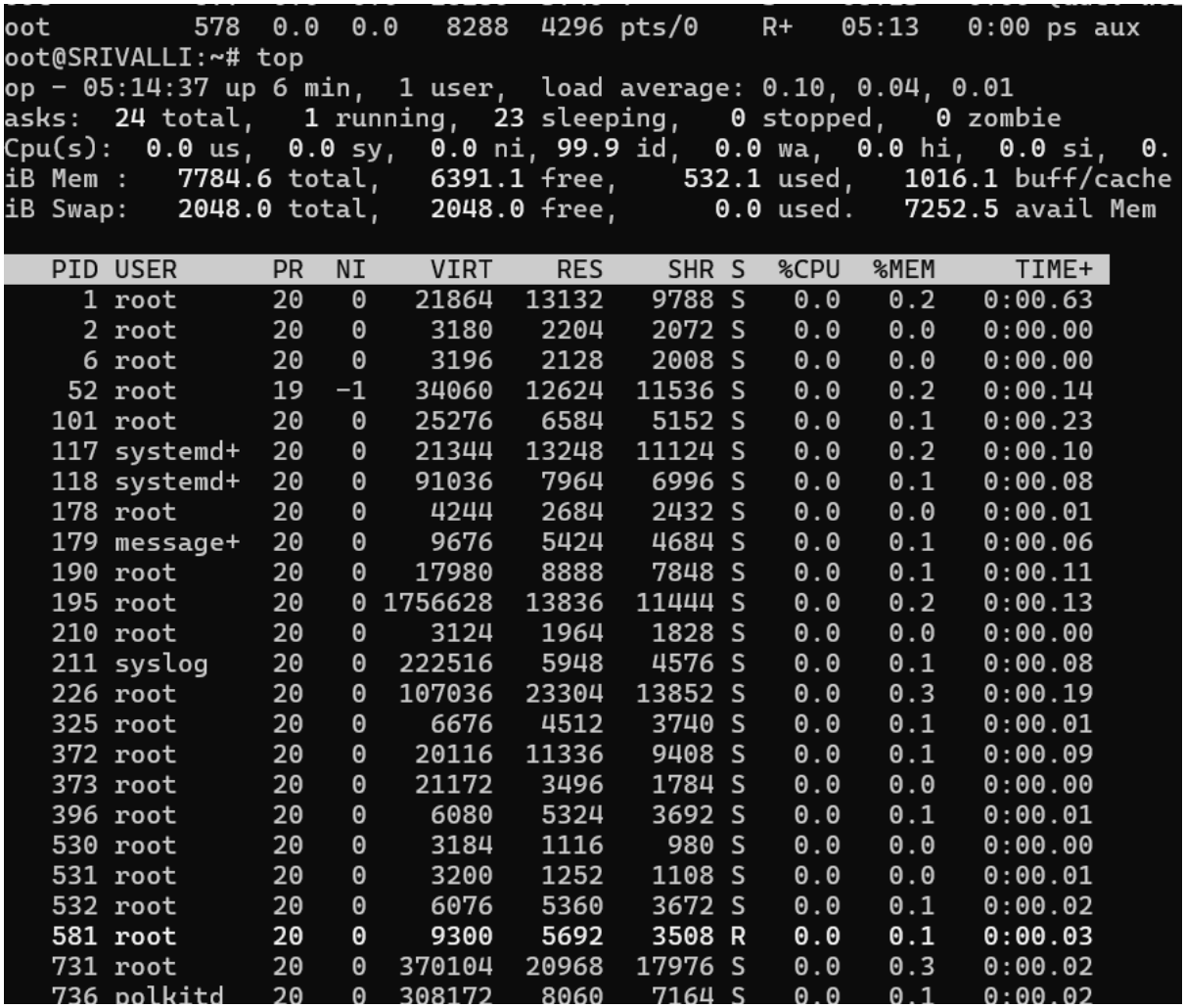
REPORT:

Title: Command Execution using fork() and exec()

The program was developed to create a parent and child process using fork(). The child process executes the Linux command entered by the user using exec(). The parent process waits for the child using wait(). The PID and PPID of both processes were also displayed.

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
root@SRIVALLI:~# uname
Linux
root@SRIVALLI:~# uname -a
Linux SRIVALLI 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu
Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
root@SRIVALLI:~# lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
  Model name:                12th Gen Intel(R) Core(TM) i5-1235U
    CPU family:              6
    Model:                   154
    Thread(s) per core:      2
    Core(s) per socket:      6
    Socket(s):               1
    Stepping:                4
    BogoMIPS:                4991.99
    Flags:                   fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_l1d arch_capabilities
```



REPORT:

Title: Hardware Resources and OS Services

In this task, Linux commands like uname, lscpu, ps, and top were used to check system information, CPU details, running processes, and resource usage. This helped us understand how the operating system manages CPU, memory, storage, and other hardware resources.